

SIMATS ENGINEERING

ASSESSMENT TOOL 3

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COURSE CODE & NAME CSA16/Data Warehousing and Data Mining

Q1. Dataset: Customer Data

Customer	Income (₹)	Spending Score
C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:

Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability.

Given Data

Customer	Income (₹)	Spending Score
C1	30,000	60
C2	60,000	40
C3	50,000	70
C4	28,000	65
C5	65,000	35

1. Data Preparation

The two features used are:

- **Income**
- **Spending Score**

Income has values in thousands, while spending score ranges only from 35–70. Therefore, **normalization is required** before clustering.

Using Min-Max normalization:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

This prevents income from dominating the distance calculation.

2. K-Means Clustering

K-Means is a **centroid-based clustering method**.

For this dataset, we can choose: K=2

The algorithm:

1. Selects two centroids.
2. Assigns customers to the nearest centroid.
3. Recalculates the centroids.
4. Repeats until the clusters become stable.

A suitable segmentation is:

Cluster 1: Higher Spending

Evidence:

C1, C3, C4

- C1 → ₹30,000, score 60
- C3 → ₹50,000, score 70
- C4 → ₹28,000, score 65

These customers have relatively high spending scores.

Cluster 2: Higher Income, Lower Spending

Evidence:

C2, C5

- C2 → ₹60,000, score 40
- C5 → ₹65,000, score 35

These customers have higher incomes but lower spending scores.

3. Hierarchical Clustering

Hierarchical clustering starts with every customer as an individual cluster:

$\{C1\}, \{C2\}, \{C3\}, \{C4\}, \{C5\}$

Similar customers are progressively merged. The process is represented using a **dendrogram**.

If the dendrogram is cut to obtain two clusters, a reasonable grouping is: $\{C1, C3, C4\}$ and $\{C2, C5\}$

Thus, both methods can identify similar customer segments for this small dataset.

4. Comparison of K-Means and Hierarchical Clustering

Criteria	K-Means	Hierarchical
Basic approach	Centroid-based	Tree-based
Number of clusters	Usually selected beforehand	Can be selected using dendrogram
Speed	Faster	Generally slower
Scalability	High	Lower
Visualization	No hierarchy	Dendrogram
Large datasets	More suitable	Less suitable
Main advantage	Efficient clustering	Shows relationships between clusters

5. Performance and Scalability

K-Means is generally faster and more suitable for large datasets. If a company has thousands or millions of customers, K-Means can efficiently create customer segments.

Hierarchical clustering requires more computation and memory. Therefore, it is more suitable for small or medium datasets where understanding the relationships between customers is important.

6. Critical Analysis

For this dataset, hierarchical clustering provides more information about **how customers are related at different levels**, while K-Means directly provides a fixed number of groups.

Therefore:

- For **large-scale customer segmentation** → **K-Means is better**.
- For **exploring customer relationships** → **Hierarchical clustering is better**.

The choice also depends on the business requirement, dataset size, and desired number of clusters.

7. Marketing Interpretation

Cluster 1: C1, C3, C4

These customers have relatively higher spending scores.

The company can use:

- Loyalty rewards
- Personalized discounts
- Premium product recommendations

Cluster 2: C2, C5

These customers have high income but lower spending scores.

The company can use:

- Personalized offers
- Special promotions
- Product recommendations

to encourage higher spending.

Conclusion

Both K-Means and hierarchical clustering can produce meaningful customer segments from the given data. The evidence suggests two useful groups: K-Means is preferred for large datasets because it is faster and more scalable, whereas hierarchical clustering is useful for smaller datasets because the dendrogram provides deeper insight into customer relationships. Thus, the best method depends on the dataset size and business objective.

Q2. Dataset: Document Data

Document	Word 1	Word 2	Word 3
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset. (BL4 – 1 Mark)

Given Data

Document	Word 1	Word 2	Word 3
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

1. K-Means Clustering

Assume: K=2

K-Means assigns each document to the **nearest cluster centroid**.

A suitable grouping is:

Cluster 1 – Word 2 Dominant

{D1,D2,D5}

D1, D2 and D5 have relatively high Word 2 values.

Cluster 2 – Word 1 Dominant

{D3,D4}

D3 and D4 have very high Word 1 values.

2. Limitation of K-Means

K-Means uses **hard clustering**.

Therefore, each document belongs to only one cluster.

For example, D5 may have characteristics of different topics, but K-Means must assign it to only one cluster.

Thus, K-Means is less suitable when documents have **overlapping topics**.

3. EM Algorithm

EM (Expectation-Maximization) is a **soft clustering** method.

Instead of assigning a document completely to one cluster, EM gives a probability of belonging to each cluster.

For example: $P(C1)=0.70, P(C2)=0.30$

This means the document is more likely to belong to C1 but still has some relationship with C2.

EM has two main steps:

E-Step: Calculate the probability of each document belonging to each cluster.

M-Step: Update the cluster parameters using these probabilities.

These steps are repeated until the model converges.

4. EM Interpretation

- **D1 and D2:** Strongly related to the Word 2 cluster.
- **D3 and D4:** Strongly related to the Word 1 cluster.
- **D5:** Has characteristics that may overlap with other groups, so EM can give it partial membership.

Exact probabilities cannot be calculated because the question does not provide initial cluster parameters.

5. K-Means vs EM

Feature	K-Means	EM
Clustering	Hard	Soft

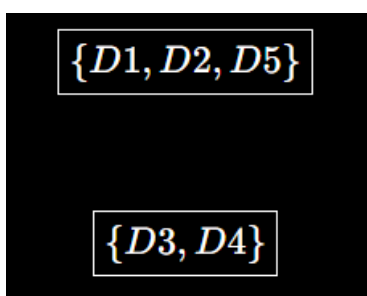
Feature	K-Means	EM
Assignment	One cluster	Probability-based
Overlapping clusters	Limited	Better
Speed	Generally faster	Generally slower
Uncertainty	Not represented	Represented
Best suited for	Clearly separated data	Overlapping data

6. Advantages of EM

1. Handles overlapping topics.
2. Provides membership probabilities.
3. Represents uncertainty.
4. Gives more detailed information about documents.
5. Better represents documents containing multiple topics.

7. Conclusion

K-Means produces:



and

It is simple and fast but uses **hard assignment**. EM is more suitable for overlapping document clusters because it provides probability-based membership. Therefore:

K-Means → Simple and fast

EM → Better for overlapping clusters

Hence, EM is preferred when documents can belong to multiple topics, while K-Means is suitable when the clusters are clearly separated.

Q3. Dataset: Geographic Data

Point	Latitude	Longitude
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question:

Compare Euclidean and Manhattan distance metrics and analyse their impact on clustering results. (BL4 – 1 Mark)

Given Data

Point	Latitude	Longitude
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

1. Euclidean Distance

Euclidean distance measures the **straight-line distance** between two points.

Formula:

$$D_E = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

For example, between P1 and P2:

$$\begin{aligned} D_E &= \sqrt{(13.10 - 13.08)^2 + (80.25 - 80.27)^2} \\ &= \sqrt{0.02^2 + (-0.02)^2} \\ &= \sqrt{0.0008} \\ D_E &= 0.0283 \end{aligned}$$

2. Manhattan Distance

Manhattan distance calculates the total absolute difference along each dimension.

Formula:

$$D_M = |x_2 - x_1| + |y_2 - y_1|$$

For P1 and P2:

$$\begin{aligned} D_M &= |13.10 - 13.08| + |80.25 - 80.27| \\ &= 0.02 + 0.02 \end{aligned}$$

$$D_M = 0.04$$

3. Distance Comparison

Some pairwise distances are:

Pair	Euclidean	Manhattan
P1-P2	0.0283	0.04
P1-P4	0.0424	0.06
P3-P5	0.0361	0.05
P1-P3	0.1208	0.16
P2-P3	0.1334	0.16

Both metrics identify **P1-P2** and **P3-P5** as very close pairs.

4. Effect on Clustering

Using Euclidean Distance

Euclidean distance measures direct spatial closeness.

A possible grouping is: {P1,P2,P4}

And {P3,P5}

P1, P2 and P4 are relatively close to one another, while P3 and P5 are also close.

Using Manhattan Distance

Manhattan distance adds the latitude and longitude differences separately.

It can produce slightly different distances and therefore may change the exact cluster boundaries, especially when points are close to each other.

For this dataset, the main groups are still likely to remain similar: : {P1,P2,P4}

And {P3,P5}

5. Difference Between the Two Metrics

Feature	Euclidean	Manhattan
Distance type	Straight-line	Sum of coordinate differences
Formula	$\sqrt{\Delta x^2 + \Delta y^2}$	(
Path	Direct	Axis-by-axis
Effect	Smooth distance measurement	More sensitive to coordinate-wise differences
Suitable for	Direct geographic/spatial similarity	Grid or road-like movement

6. Impact on Clustering Results

The choice of distance metric affects how **similarity between points is calculated**.

- **Euclidean distance** is generally suitable when direct spatial distance is important.
- **Manhattan distance** can be useful when movement follows separate latitude/longitude directions or a grid-like structure.
- Different distances can change which points are considered nearest.
- Therefore, the selected metric can influence cluster boundaries and the final grouping.

For this dataset, because the points are geographically close and form clear pairs/groups, **both methods give broadly similar segmentation**, although their numerical distances are different.

7. Critical Analysis

For actual geographic locations, latitude and longitude are angular coordinates, so for accurate real-world distances a **geodesic/Haversine distance** is generally preferable, especially over larger areas.

For this small dataset covering a relatively small geographic region, Euclidean distance can be used as a simple approximation after appropriate preprocessing.

Conclusion

Euclidean distance calculates the **straight-line distance**, while Manhattan distance calculates the **sum of absolute differences**.

For example:

$$P1 - P2 : \textit{Euclidean} = 0.0283$$

$$P1 - P2 : \textit{Manhattan} = 0.04$$

Both methods identify similar groups in this dataset, such as **P1, P2, P4** and **P3, P5**. However, the choice of distance metric can affect cluster formation when points are closer or more irregularly distributed.

Therefore, **Euclidean is preferable for direct spatial similarity, while Manhattan is useful for axis-based or grid-like distance calculations**.

Q4. Dataset: Retail Products

Product	Sales(₹)	Frequency
P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset.
(BL3 – 1 Mark)

Given Data

Product	Sales (₹)	Frequency
P1	50,000	20
P2	30,000	10
P3	52,000	22
P4	25,000	8
P5	32,000	12

1. Understand the Data

The two features used for clustering are:

- Sales
- Purchase Frequency

Since sales values are much larger than frequency values, **normalization should be applied** before calculating distances.

Using Min-Max normalization:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

This ensures that both features contribute fairly to clustering.

2. Agglomerative Clustering

Agglomerative clustering is a **bottom-up** approach.

Initially, every product is treated as a separate cluster: $\{P1\}, \{P2\}, \{P3\}, \{P4\}, \{P5\}$

The most similar clusters are progressively merged.

Step-by-step

- P1 and P3 are very similar because both have high sales and high frequency.
- P2, P4 and P5 have relatively lower sales and frequency.

A suitable final grouping is:

Cluster 1 – High-Performing Products

$\{P1, P3\}$

- P1 → ₹50,000, frequency 20
- P3 → ₹52,000, frequency 22

Cluster 2 – Lower-Performing Products

{P2,P4,P5}

- P2 → ₹30,000, frequency 10
- P4 → ₹25,000, frequency 8
- P5 → ₹32,000, frequency 12

3. Divisive Clustering

Divisive clustering is a **top-down** approach.

Initially, all products are placed into one cluster: { P1 }, { P2 }, { P3 }, { P4 }, { P5 }

The algorithm then divides the cluster into smaller groups.

First division

The products can be divided into: { P1, P3 }

and { P2, P4, P5 }

Further divisions can be performed if more detailed product segments are required.

4. Comparison of Agglomerative and Divisive Clustering

Feature	Agglomerative	Divisive
Approach	Bottom-up	Top-down
Starting point	Individual products	One large cluster
Process	Merges clusters	Splits clusters
Dendrogram	Shows merging	Shows splitting
Complexity	Generally easier to implement	Generally more complex
Suitable for	Small/medium datasets	Exploratory hierarchical analysis
Main advantage	Simple and intuitive	Finds broad groups first

5. Suitability for This Dataset

This dataset contains only **5 products**, so both approaches can be used.

Agglomerative is suitable because:

- The dataset is small.
- Similar products can be easily merged.
- It provides a clear hierarchy of product similarity.
- It can help identify high- and low-performing products.

Divisive is suitable because:

- It starts with all products together.
- It can first separate high-performing products from lower-performing products.
- It is useful when the company wants to understand broad groups first.

6. Business Interpretation

Cluster 1 – High-Performing Products [P1,P3]

These products have high sales and high purchase frequency.

The retailer can:

- Maintain sufficient inventory.
- Promote these products.
- Give them more shelf/display space.
- Use them as popular products in marketing campaigns.

Cluster 2 – Lower-Performing Products [P2,P4,P5]

These products have lower sales and frequency.

The retailer can:

- Provide discounts.
- Create bundle offers.
- Study customer preferences.
- Improve product promotion.

7. Critical Comparison

Agglomerative clustering is generally easier to understand for this dataset because it gradually combines similar products. It is useful when the retailer wants to see **which products are similar step by step**.

Divisive clustering starts with all products and separates them into groups. It is useful when the retailer first wants to identify **major product categories** and then divide them further.

For a very large dataset, hierarchical methods can become computationally expensive. For this small dataset, however, both approaches are practical.

Conclusion

Both **agglomerative and divisive clustering** can be applied to the given retail dataset.

A meaningful segmentation is:

$\{P1, P3\} \rightarrow \text{High sales, High frequency}$

$\{P2, P4, P5\} \rightarrow \text{Lower sales, Lower frequency}$

Agglomerative clustering builds clusters from individual products by merging similar products, while **divisive clustering** starts with all products and repeatedly splits them.

For this small dataset, **agglomerative clustering is slightly more convenient and intuitive**, while divisive clustering is useful when the retailer wants to study broad groups first. Both can support **inventory planning, product promotion, sales analysis, and marketing decisions**.

Item	Price (₹)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:

Compare centroid-based and density-based clustering techniques and analyse their effectiveness on mixed data types.

Given Data

Item	Price (₹)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

The dataset contains **mixed attributes**:

- Price → Numerical
- Category → Categorical
- Rating → Numerical

1. Data Preparation

Centroid-based methods such as K-Means require numerical data.

Therefore, the categorical **Category** attribute must first be converted into numerical form using **one-hot encoding**.

Item	Price	Category A	Category B	Rating
I1	500	1	0	4.5
I2	300	0	1	3.8
I3	550	1	0	4.7
I4	250	0	1	3.5
I5	520	1	0	4.6

Price should also be **normalized** because its numerical scale is much larger than rating.

2. Centroid-Based Clustering

K-Means is a common centroid-based clustering technique.

Assume: $K=2$

The algorithm:

1. Selects two centroids.
2. Calculates the distance of each item from the centroids.

3. Assigns each item to the nearest centroid.
4. Recalculates the centroids.
5. Repeats until the clusters become stable.

3. Centroid-Based Cluster Formation

A natural grouping is:

Cluster 1 – Category A / High Price and Rating{I1,I3,I5}

Evidence:

- I1 → ₹500, A, 4.5
- I3 → ₹550, A, 4.7
- I5 → ₹520, A, 4.6

These items have similar categories, prices, and ratings.

Cluster 2 – Category B / Lower Price and Rating{I2,I4}

Evidence:

- I2 → ₹300, B, 3.8
- I4 → ₹250, B, 3.5

These items also have similar characteristics.

4. Density-Based Clustering

DBSCAN is a common density-based clustering technique.

It groups data points that are located in **dense regions** and identifies isolated points as noise.

DBSCAN mainly uses two parameters:

- **ε (Epsilon):** Neighborhood radius
- **MinPts:** Minimum number of points required to form a dense region

For mixed data, a suitable distance measure or preprocessing method is required because Category is categorical.

5. Density-Based Interpretation

The dataset contains two relatively compact groups: {I1,I3,I5}

And {I2,I4}

The first group contains Category A items with high prices and ratings, while the second contains Category B items with lower prices and ratings.

However, the exact DBSCAN result depends on the selected **ε and MinPts** values.

6. Comparison

Feature	Centroid-Based (K-Means)	Density-Based (DBSCAN)
Basic idea	Groups around centroids	Groups dense regions
Number of clusters	Usually specified	Not specified directly
Shape of clusters	Works best with compact clusters	Can find irregular shapes
Noise detection	Poor	Good
Mixed data	Needs encoding	Needs suitable mixed-data distance
Speed	Generally fast	Can be efficient
Main limitation	Sensitive to initialization and outliers	Sensitive to ε and MinPts

7. Effectiveness on This Dataset

Centroid-Based

K-Means is effective because the data naturally forms two compact groups: {I1,I3,I5}

And {I2,I4}

It is simple and easy to interpret.

Density-Based

DBSCAN can also identify these groups if suitable ϵ and MinPts values are selected. Its advantage is that it can identify **outliers or isolated products**.

However, with only five items, choosing suitable density parameters can be difficult.

8. Critical Analysis

For this particular dataset, **K-Means is more straightforward** because the groups are clearly separated by category, price, and rating.

DBSCAN becomes more useful when the dataset contains:

- Irregularly shaped groups
- Large numbers of items
- Noise or outliers
- Unknown number of clusters

For mixed attributes, both methods require appropriate preprocessing or a distance measure that can handle numerical and categorical variables.

9. Business Interpretation

Cluster 1 – Premium Products {I1,I3,I5}

These products have higher prices and ratings and belong to Category A.

The retailer can:

- Promote them as premium products.
- Maintain adequate stock.
- Use them in premium marketing campaigns.

Cluster 2 – Budget Products {I2,I4}

These products have lower prices and ratings and belong to Category B.

The retailer can:

- Offer discounts.
- Create bundle offers.
- Improve product promotion.

X10. Conclusion

Both **centroid-based and density-based clustering** can be used for this mixed dataset after suitable preprocessing.

A meaningful segmentation is:

$\{I1, I3, I5\} \rightarrow \text{Premium Category A}$

$\{I2, I4\} \rightarrow \text{Lower Price Category B}$

K-Means is more suitable for this small, clearly separated dataset because it is simple and produces easily interpretable clusters.

DBSCAN is more useful when clusters have irregular shapes or when detecting outliers is important.

Therefore, for this dataset, **centroid-based clustering is the more straightforward choice**, while density-based clustering provides advantages for more complex datasets.